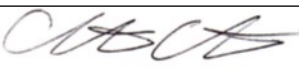


Guideline Title: Pediatric General Management

Guideline Number: GUID-3203-PED001

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To identify historical data relevant to the care of the pediatric patient population.
- II. **DEFINITIONS:**
 - a. Neonate: birth to 28 days
 - b. Infant: 1 month to 1 year
 - c. Child: 1 year to 17 years (except in the case of trauma < 16 years old)
 - d. Adult > or = to 18 years old
 - e. NOTE: The American Heart Association defines the pediatric patient as any prepubescent patient (e.g., absence of breast development in the female or pubic or axilla hair in the male).
- III. **DEPARTMENT GUIDELINE:**
 - a. Obtain a good **HISTORY**:
 - i. Obtain a concise history; identify appropriate informant (patient, family, practitioner).
 - ii. Gather past medical history: congenital problems, allergies, childhood illnesses or recent exposure to illness, current immunization status, medications/OTCs, & previous hospitalizations/surgeries.
 - iii. For infants: note if full term; breast/bottle fed, complications during pregnancy or any relevant NICU history.
 - iv. Obtain history of present illness: time of onset, initial presenting signs and symptoms, recent illnesses of child or other close family member, recent fluid intake and output, and accurate weight prior to illness and current weight and patient's response to therapy.
 - v. Obtain history of treatment prior to arrival and pt's response to therapy.
 - vi. For inter-facility transfers obtain: vital signs including temperature, assessment parameters, radiographic studies, laboratory results
 1. Transport should not be delayed if information is not readily available
 - b. Obtain a good **ASSESSMENT**:
 - i. Airway patency should be immediately assessed, and emergent intervention should be performed. Key findings indicating significant respiratory compromise on exam are not limited to, but could include dysphagia, stridor, snoring, gurgling, tripod posturing.
 - ii. Assess the patient's respiratory status:

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1. Rate: Pediatric patients have a higher respiratory rate at baseline due to a higher basal metabolic rate. A respiratory rate consistently > 60 breaths/minute in a child of any age is abnormal and is considered a “red flag”.
2. Oxygenation: Appropriate oxygenation can be assessed by evaluating skin and mucous membrane color, mental status, muscular tone, oxygen saturation, and heart rate. The response of the pediatric cardiopulmonary system to hypoxia is bradycardia.
3. Ventilation: Assessment of ventilation begins with a thorough auscultation of breath sounds. ETCO₂ may be assessed during transport period. Baseline ABG determinations may be available on interfacility transports.
4. Work of breathing: It should be evaluated examining for the presence or absence of retractions, nasal flaring, grunting, cyanosis, or use of other accessory muscles.
- iii. Assess circulation:
 1. Identify heart rate as appropriate for age.
 2. If external hemorrhage is identified, bleeding should be controlled immediately with well-aimed direct pressure or other indicated techniques.
 3. Presence or absence of peripheral and central pulses, capillary refill time, skin color & temperature, and hypovolemia (skin turgor, lack of tearing, dry mucous membranes, sunken fontanels and eyes in infants).
- iv. Assess central nervous system (CNS) perfusion by level of consciousness & muscle tone.
- v. Kidney perfusion can be assessed by urine output (in the absence of renal disease). Adequate urinary output in an infant is 1-2 ml/kg/hr and in an older child and adolescent is 0.5-1 ml/kg/hr.
- c. Obtain patient weight or use a Broselow tape to estimate patient weight.
- d. For interfacility pediatric transfers, call receiving physician prior to departure to update them on current status and to discuss plan of care.
- e. **INTERVENTIONS:**
 - i. If patient has adequate spontaneous respirations, administer supplemental oxygen to maintain SpO₂ > 93%.
 - ii. Connect to cardiac monitor, NIBP, pulse oximeter and ETCO₂
 - iii. Obtain a set of vital signs with serial assessment.
 - iv. Establish an IV(s) with age appropriate IV size, rate and solutions.
 - v. If unable to obtain IV access, IO access should be obtained without further delay when indicated.
 - vi. If the patient has a GCS of < 9 and/or has a deteriorating level of consciousness, consider advanced airway management.
 - vii. Nauseated patients may be treated per *GUID3203-G004 – Nausea/Vomiting/Motion Sickness*
 - viii. Treat pain and anxiety per *GUID3203-G005 - Analgesia and Sedation Management* or *GUID3203-G006 Anxiety- Combative Patient Management*.
 - ix. Consider placement of an oro/nasogastric tube for a child with gastric distention per *GUID3203-PR010 Gastric Tube Insertion*.
 - x. For interfacility or long-range transports only, consider maintenance fluids if not already initiated. Preferred maintenance fluid is D5NS:
 1. 4 mL/kg/hr IV for the first 10 kg.

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2. 2 mL/kg/hr IV for the second 10 kg.
3. 1 mL/kg/hr IV for each kg of body weight above 20 kg.
4. **Pediatric Patient - Maintenance Fluids Calculation Table**

Patient Weight in kg	mL per hour IV	Patient Weight in kg	mL per hour IV
5	20	30	70
10	40	35	75
15	50	40	80
20	60	45	85
25	65	50	90

SPECIAL SITUATION - Suspected Abuse:

- xi. For pediatric patients suffering traumatic injury, the provider must consider physical abuse as part of the differential diagnosis. However, other types of abuse, such as neglect or emotional abuse, do occur. There may be secondary medical findings, or the suspicion of abuse may be incidental to the chief complaint. For example, an infant being transported with wound sepsis after an inflicted burn, or a child with failure to thrive and a suspicious social situation.
- xii. The State of Florida requires medical and pre-hospital personnel to report instances of suspected abuse to local child protection agencies. It may be required that *each* mandated provider generate a report after treating and potentially identifying a suspected abuse victim. Each program must be familiar with the reporting process defined by their area. The medical transport team must obtain a careful history to compare reported mechanism of injury with observed or diagnosed injury patterns. Inconsistency here or the “story not matching up,” should lead the team to suspect potential abuse. This should be reported not only to local child protection entities, but also to the receiving facility and be documented in the patient care report.

IV. DOCUMENTATION:

- a. EMS Charts

V. REFERENCES:

- a. Pediatric Advanced Life Support Manual, 2020. American Heart Association.